Narrative Selection and Societal Endurance

A Stage 1 Registered Report protocol testing whether the moral content of cultural traditions predicts the survival of the societies that carry them

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Abstract

Genetic inheritance is constrained by the Weismann barrier: what an organism learns in its lifetime cannot be written into its germ line. Humans possess a second inheritance system, language, through which learned information is transmitted directly across generations in the form of narratives, laws and archetypes. If narratives are subject to selection against the environment of human social life, then the moral content of the narratives a society carries should predict that society's endurance, independently of the material drivers of power. Previous tests of related claims (the "Big Gods" hypothesis) measured belief in supernatural agents rather than narrative content, measured growth in social complexity rather than endurance, and did not distinguish traditions that persist because they serve human social life from traditions that persist because their carriers won wars or out-reproduced rivals. We propose three pre-registered studies: (1) a historical survival analysis of polities in the Seshat Global History Databank, with narrative content coded blind to outcome; (2) a cross-cultural analysis of the Berezkin folklore catalogue (about 1,000 societies) linked to historical and contemporary outcomes; and (3) a multi-site randomized experiment testing whether exposure to specific narrative content causally increases cooperation with strangers. We specify falsification criteria, equivalence bounds, competing-risk tests that separate internal collapse from conquest, and a test of temporal invariance. We also state explicitly which conclusions no empirical result can license.

1. Introduction

1.1 Theoretical claim

Evolution by natural selection operates on replicators whose phenotypic effects differ in persistence relative to an environment. In genetic evolution the Weismann barrier (Weismann, 1893) prevents acquired information from entering the germ line, so populations "learn" only through differential death and reproduction across generations. Dual inheritance theory (Boyd and Richerson, 1985) holds that culture constitutes a second inheritance system with its own variation, transmission and selection. Narrative transmission crosses the Weismann barrier: an individual's learned lesson can be encoded and inherited without genetic change. Two properties follow. Variation is guided rather than blind, because transmitters revise content

toward what has worked; and selection operates on content against the environment of human social life.
Narrative selection hypothesis (NSH): the moral content of a society's narrative tradition is a selectable trait, and content that encodes cooperation-supporting norms raises the expected endurance of the society carrying it, after accounting for the material determinants of power.
The circularity objection ("the traditions that survive are those that survive") parallels the classic tautology objection to "survival of the fittest". It is resolved here as in biology, by the propensity interpretation of fitness (Mills and Beatty, 1979): the trait (narrative content) and the outcome (endurance) are measured independently, and the trait is used to predict the outcome before the outcome is observed by coders.

Peterson (2013) gives a psychological account of the same mechanism. He distinguishes three forms of meaning: the meaning of the known, ordered world, structured by motivation-action-perception (MAP) schemas nested within social dominance hierarchies; the meaning of the unknown and anomalous; and the meaning of their conjunction, found in voluntary exploration of the unknown. On his account, the great myths are not theories of objective existence but dramatic representations of patterns of action that proved adaptive across generations, abstracted from observed behaviour before they could be stated as explicit rules. That is the content NSH proposes to measure. Peterson also derives a meta-moral principle from repeated play: the most successful player is not the one who wins a single game but the one others keep inviting to play, so a strategy that trades future invitations for present victory loses over time; following de Waal (1982), he notes that even among chimpanzees rule by strength alone is unstable. This principle corresponds formally to partner-choice models of the evolution of morality (Barclay and Willer, 2007; Baumard, Andre and Sperber, 2013) and to the success of reciprocating strategies in iterated games (Axelrod, 1984), and it generates hypothesis H8.

1.2 Prior evidence

- **Big Gods.** Norenzayan (2013) and Norenzayan et al. (2016) argue that culturally evolved prosocial religions with moralizing, punishing deities supported cooperation, trust and fertility in large anonymous societies and helped those groups prevail in intergroup competition.
- **Economic games.** Purzycki et al. (2016) found across field sites that belief in more knowing and punishing gods predicted more impartial allocation to distant co-religionists.
- **Austronesia.** Watts et al. (2015), using phylogenetic methods, found that broad supernatural punishment, but not moralizing high gods, preceded the evolution of political complexity.
- **Seshat 2019 and its retraction.** Whitehouse et al. (2019) concluded that moralizing gods followed rather than caused increases in social complexity. Beheim et al. (2021) showed the result depended on coding missing data as absence; the paper was retracted in July 2021. A corrected analysis (Whitehouse et al., Religion, Brain and Behavior 13(2), 124-166) maintains that the largest increases in social complexity preceded Big Gods. The causal direction remains disputed.
- **Material drivers.** Turchin et al. (2022), testing 17 candidate predictors on Seshat data, found the best-supported drivers of social complexity were agricultural productivity and military technology, notably iron weapons and cavalry.
- **Folklore as data.** Michalopoulos and Xue (2021) coded thousands of motifs across

roughly 1,000 societies from Berezkin's catalogue and showed that folklore content predicts gender norms, risk attitudes and trust, including among second-generation immigrants; for example, societies whose tales depict tricksters being punished show higher trust.

- **Kinship and moral scope.** Enke (2019) showed that loose kinship structures predict universalist rather than communal moral systems; Schulz et al. (2019) linked historical exposure to the Western Church, through the dissolution of intensive kinship, to impersonal prosociality.
- Leadership and hierarchy. Boehm (1993) documented reverse dominance hierarchies, in which egalitarian foragers collectively sanction would-be dominators. Sapolsky and Share (2004) found that after aggressive males died in a tuberculosis outbreak, a baboon troop maintained a more tolerant culture, in which low-ranking males no longer showed the elevated glucocorticoid levels seen before the outbreak, through a complete turnover of adult males; however, dominance stability and the tenure of the top male did not differ from control troops. Foster et al. (2009) found that the three Gombe chimpanzee alphas of 1989 to 2003 had stable and distinct dominance styles, and the largest, least-grooming alpha held the position longest (n = 3). Cheng et al. (2013) showed that dominance and prestige are independent and both effective routes to influence in human groups, but only prestige is liked. Blanton et al. (2020), in 30 premodern states, found that good government was associated with commercial development but not with significantly longer duration (166 versus 152 years on average), and that good-government states suffered more severe collapses; in four case histories they traced the initial cause of collapse to rulers abandoning their role as moral exemplars. The evidence that humble, cooperative leadership sustains hierarchies longer is therefore mixed, and is treated here as a hypothesis to test, not a premise.
- Ruler accountability data. The current Seshat release codes 'moralizing enforcement of rulers' (whether rulers are held subject to supernatural moral sanction) for 454 polities: 298 present, 74 absent, 78 unknown, 6 transitional (audit of 16 October 2026).
- Moral universals versus gods. Curry, Mullins and Whitehouse (2019) found seven cooperative behaviours judged morally good in nearly every instance across the ethnographic records of 60 societies. Belief in a moralizing high god is far less general: in the Ethnographic Atlas (D-PLACE), 198 of 775 coded societies (25.5 percent) have one, and 498 (64.3 percent) have a high god of any kind.

1.3 Gaps this study addresses

1. **Vessel versus message.** Prior tests measure belief in supernatural agents. NSH concerns the moral content of narratives, which may be carried by supernatural belief, secular law or folklore alike.
2. **Growth versus endurance.** Prior tests predict increases in complexity. NSH predicts persistence and recovery from crisis.
3. **Truth-to-human-life versus spread.** No prior test separates content that persists because it supports social life from content that persists because its carriers conquered or out-reproduced others.
4. **Causal mechanism.** Historical correlations cannot establish that content changes behaviour. An experimental arm is required.

2. Hypotheses and falsification conditions

ID	Hypothesis	Prediction	Falsified if
H1	Content predicts endurance	Higher Cooperative Content Index (CCI) lowers the hazard of polity termination (HR per SD below 1)	90% CI for HR lies within the equivalence bounds 0.90 to 1.11, or HR is above 1
H2	Robustness to drivers of power	H1 holds after adjusting for agricultural productivity, military technology, population, social complexity, wealth proxies, geography and kinship intensity	Adjusted HR within equivalence bounds
H3	Temporal precedence	CCI at century t predicts termination hazard and crisis recovery at t+1, and CCI changes precede rather than follow complexity changes	Lagged effect null while reverse lag (complexity at t predicting CCI at t+1) is significant and larger
H4a	Internal versus external exit	CCI reduces the sub-hazard of internal collapse (fragmentation, civil war, state failure) at least as much as the sub-hazard of conquest	CCI predicts only conquest avoidance, with internal-collapse sub-hazard within equivalence bounds
H4b	Reverse transmission	When a militarily dominant group adopts narrative content from a subordinate group, the adopted content has higher CCI than the non-adopted content of either group	Adopted content does not differ from a prestige- and literacy-matched baseline
H5	Vessel independence	CCI predicts endurance	CCI effect disappears when moralizing high

ID	Hypothesis	Prediction	Falsified if
		independently of the presence of moralizing high gods, and the effect is mediated in part by the scope of trust	gods are controlled
H6	Individual-level mechanism	Random exposure to high-CCI narratives increases cooperation with strangers in incentivized games relative to matched control narratives	Effect within the equivalence bounds d = -0.15 to 0.15
H7	Temporal invariance	The CCI-endurance relation does not differ across eras (before 500 BCE, 500 BCE to 1500 CE, after 1500 CE)	Significant era-by-CCI interaction with sign reversal

H7 operationalizes the claim that these narrative truths are atemporal for humans: if the relation holds across radically different technological environments, it reflects human social life rather than a particular period.

H8 (added 16 October 2026): leadership legitimated by service outlasts leadership held by force. Formal basis: a leader extracts a share e of a coalition's surplus S produced by n members, and each member stays while (1 - e)S/n exceeds an outside option w (exit, a rival coalition, or collective sanction). Coercion raises the cost c of acting on the outside option, so the tolerable extraction rises from 1 - nw/S to 1 - n(w - c)/S. A dominance strategy therefore sustains higher extraction only while c stays high, and any shock that lowers c or S (ageing, injury, defeat, famine, a rival coalition) moves the polity past its threshold at once; a prestige strategy that keeps e below 1 - nw/S has no such threshold. The model predicts that dominance-based hierarchies need not be shorter on average, consistent with Foster et al. (2009) and Blanton et al. (2020), but should end more often through internal collapse and more violently. H8a: polities whose rulers are subject to moralizing enforcement, or whose canonical narratives present legitimate rule as service bounded by moral law (a Ruler Constraint item added to the narrative coding, 0 to 3, coded blind like the CCI), have a lower sub-hazard of internal collapse; conquest is not predicted. H8b: conditional on termination, these polities show lower crisis severity in CrisisDB (civil war, extermination, collapse). Blanton et al. (2020) report the opposite pattern in 19 cases, so H8b is a risky test. Falsified if the internal-collapse sub-hazard ratio lies within the equivalence bounds or above 1 (H8a), or if severity is equal or higher (H8b). Power: with the binary Seshat variable (80.1 percent present among known values), 505 events are needed to detect a hazard ratio of 0.6 at 90 percent power; with the 33 internal-collapse events now available the minimum detectable hazard ratio is 0.14. H8 is therefore exploratory until the coded sample grows, and no analysis linking ruler accountability to outcomes will be run before the coded sample is frozen, because an early look would forfeit

its confirmatory status. Historiographic bias is a specific threat: rulers who were overthrown tend to be remembered as tyrants, so ruler conduct is coded only from sources describing the period before the end of the reign or polity, never from accounts of its fall.

H9 (added 16 October 2026): independent convergence on cooperative moral content, not on belief in a moralizing God. H9a: in a new, phylogenetically balanced sample (one society per language family, excluding Curry et al.'s 60), each of the seven cooperative behaviours is judged good wherever it appears; falsified if any behaviour is judged bad in more than 5 percent of instances in which it is coded. H9b: belief in a moralizing high god (Ethnographic Atlas EA034) shows strong phylogenetic and spatial signal and depends on social scale, indicating spread by descent and contact rather than independent discovery; falsified if its phylogenetic signal (D statistic; Fritz and Purvis, 2010) is no stronger than that of the cooperative norms. Together, H9a and H9b distinguish convergence of moral content, which the framework predicts, from convergence of deity belief, which the data already show is partial.

3. Operational definitions

Construct	Operational definition	Source	Quality criterion
Narrative tradition	The set of canonical narratives (sacred texts, epics, legal-mythic texts, oral motifs) documented as widely transmitted within a polity during a given century	Seshat religion and ritual variables; Database of Religious History; Berezkin catalogue; primary texts	Corpus inclusion rules pre-registered; minimum two independent sources per polity-century
Cooperative Content Index (CCI)	Mean standardized score across the seven Morality-as-Cooperation domains (family, group loyalty, reciprocity, heroism, deference, fairness, property; Curry et al., 2019), each coded 0 to 3 for endorsement in the narrative corpus, multiplied by a moral scope weight	Coded by protocol in Section 5	Krippendorff alpha at least 0.80 per domain; confirmatory factor analysis with measurement invariance across macro-regions (configural and metric)
Moral scope	Degree to which norms are extended beyond kin and in-group: 1 = kin only, 2 = co-ethnics or	Coded by protocol	Alpha at least 0.80; validated against Enke (2019) universalism scores where societies overlap

Construct	Operational definition	Source	Quality criterion
	co-religionists, 3 = strangers including out-group members		(target r at least 0.4)
Polity	Seshat polity unit: an independent political entity with a defined territory and centre of authority	Seshat Global History Databank	As defined by Seshat codebook
Endurance	Time from polity start to termination, in years	Seshat polity dates	Date uncertainty handled by drawing from Seshat date ranges in 1,000 imputations
Termination type	(a) internal collapse: fragmentation, state failure or civil war ending central authority; (b) conquest by an external polity, with absorption, population replacement and secession coded as separate categories under the Termination-Type Codebook; (c) transformation: succession with territorial and institutional continuity (treated as censoring)	Seshat succession variables; CrisisDB; coded by protocol	Alpha at least 0.80 on termination type
Crisis recovery	For each crisis episode, time to restoration of pre-crisis territory or population, or permanent loss, plus severity of consequences	CrisisDB (Seshat Crisis Database, 168 historical cases; Hoyer et al., Social Science History)	CrisisDB coding standards
Impersonal trust	Historical: presence of impersonal institutions (written contracts,	Seshat institutional variables; Falk et al. (2018); WVS	Pre-registered linkage of ethnic groups to survey respondents

Construct	Operational definition	Source	Quality criterion
	courts open to strangers, long-distance credit). Contemporary: Global Preferences Survey trust item and World Values Survey generalized trust of descendant populations		following Michalopoulos and Xue (2021)
Moralizing high gods	Seshat moralizing supernatural punishment and moralizing high god variables	Seshat; Pulotu for Austronesia	Missing treated as missing, never as absent
Agricultural productivity	Seshat carrying-capacity estimates	Seshat Agricultural Productivity project	As published
Military technology	Seshat military technology index (including iron weapons and cavalry)	Seshat; Turchin et al. (2022)	As published
Social complexity	First principal component of Seshat complexity characteristics	Turchin et al. (2018)	As published
Population and fertility proxy	Log polity population and century-on-century growth rate	Seshat	Imputation for missing values
Kinship intensity	Kinship intensity index	Ethnographic Atlas via D-PLACE; Schulz et al. (2019)	As published
Geography and climate	Latitude, ruggedness, river access, distance to coast, drought and temperature anomalies	Standard GIS sources; paleoclimate reconstructions	Pre-registered sources
Writing and literacy	Seshat information complexity variables	Seshat	Included because text survival depends on writing

4. Studies and samples

Study 1: Historical survival panel

Unit: polity-century within Seshat Natural Geographic Areas. Inclusion: all polities with at least two independent sources for narrative content and recorded start and end dates, including extinct polities, which removes survivorship bias at the polity level. The narrative coding is performed on the content of the tradition, so polities that left few texts are retained with appropriate missingness modelling rather than dropped.

Study 2: Cross-cultural folklore

Unit: ethnolinguistic group in Berezkin's catalogue linked to the Ethnographic Atlas (D-PLACE). Outcomes: historical political continuity, presence of impersonal institutions, and contemporary trust of descendant populations, including second-generation immigrants to separate cultural transmission from current institutions. Motif summaries are coded for CCI by the protocol in Section 5.

Study 3: Multi-site randomized experiment

Design: participants in at least eight countries spanning distinct cultural regions are randomized to read one of a set of traditional narratives high in CCI (with emphasis on moral scope toward strangers) or a matched control narrative from the same tradition, equated for length, readability, emotional valence, arousal and familiarity (pre-tested). Primary outcome: amount sent to an anonymous stranger in an incentivized trust game; secondary outcomes: public goods contribution and the Purzycki et al. (2016) random allocation game with a distant stranger. Manipulation check and attention checks are pre-registered; exclusions are fixed in advance.

5. Coding protocol

1. **Codebook.** Definitions, anchors and examples for each Morality-as-Cooperation domain and moral scope level are written and frozen before coding begins, and deposited with the Stage 1 protocol.
2. **Blinding.** Coders receive narrative passages and motif summaries with polity names, place names, dates and identifying proper nouns redacted. Coders never see outcome data. A blinding check asks coders to guess the polity; if guess accuracy exceeds chance by more than 10 percentage points, those items are recoded by coders from unrelated specialisms.
3. **Coders.** Two independent trained human coders per item, with a third adjudicator for disagreements. A large language model coder, run with a frozen prompt and version, is used only if it achieves agreement with adjudicated human codes at or above human-human agreement on a held-out validation set of at least 500 items.
4. **Reliability.** Krippendorff alpha at least 0.80 per domain before full coding proceeds; items below threshold are recoded after codebook clarification, which is logged as a deviation.
5. **Pilot.** A pilot on 5% of items, drawn at random, is used only to estimate reliability and is excluded from hypothesis tests.

6. Analysis plan

6.1 Primary model (Study 1, H1 and H2)

Cox proportional hazards model with time-varying covariates measured per century, shared frailty by Natural Geographic Area, and a spatial covariance term based on great-circle distance. The CCI is standardized. Proportional hazards are tested with Schoenfeld residuals; if violated, a discrete-time logistic hazard model with century dummies is the pre-registered alternative.

6.2 Competing risks (H4a)

Fine and Gray (1999) sub-distribution hazards for internal collapse and conquest, with transformation as censoring. The key quantity is the CCI sub-hazard ratio for internal collapse.

6.3 Temporal precedence (H3)

Cross-lagged panel models within Natural Geographic Area using one-century lags of CCI, social complexity and military technology, plus a dynamic regression framework following Turchin et al. (2022) in which CCI is added to the published best-supported model.

6.4 Reverse transmission (H4b)

Pre-registered list of episodes in which a militarily dominant group came to rule a subordinate population with a distinct narrative tradition. For each episode, narrative elements of both groups are coded and the elements that persisted in the dominant group a century later are identified. Conditional logistic regression tests whether CCI predicts adoption, controlling for the prestige and literacy of the source tradition.

6.5 Vessel independence and mediation (H5)

Models from 6.1 with the moralizing high gods variable added, and causal mediation analysis with impersonal trust institutions as mediator. Sensitivity to unmeasured mediator-outcome confounding is reported.

6.6 Missing data

Following the lesson of Beheim et al. (2021), unknown values are never treated as absent. Multiple imputation by chained equations (van Buuren and Groothuis-Oudshoorn, 2011) with 50 imputations, conditioning on region, period and related variables. Pre-registered sensitivity analyses: complete-case analysis, imputation with region-by-period interactions, and a tipping-point analysis reporting how far imputed values must shift to overturn each conclusion.

6.7 Non-independence (Galton's problem)

Societies share ancestry and contact, so observations are not independent. Models include

spatial random effects, phylogenetic random effects using Glottolog language trees, and macro-region fixed effects. Results are reported with and without each correction.

6.8 Study 2 and Study 3 models

Study 2: linear and logistic regressions of outcomes on CCI with the controls in Section 3, spatial and phylogenetic random effects, and the epidemiological second-generation immigrant design for contemporary trust. Study 3: mixed-effects linear model of amount sent with condition as fixed effect and site and narrative item as random effects.

6.9 Inference criteria

- Primary hypotheses H1 to H7 are tested with Holm-Bonferroni correction at family-wise alpha 0.05.
- Smallest effect sizes of interest: HR per SD of 0.90 (or its reciprocal 1.11) for hazard models; d = 0.15 for the experiment. Equivalence is tested with two one-sided tests (Lakens, 2017), so null results are interpretable as evidence of absence rather than absence of evidence.
- A hypothesis is supported only if the corrected test rejects the null and the estimate is robust across the sensitivity analyses in 6.6 and 6.7. It is falsified if the equivalence test succeeds.

6.10 Power

For a continuous covariate in a Cox model, the required number of events is $D = (z_{1-alpha/2} + z_{1-beta})^2 / [(\ln HR)^2 \times (1 - R^2)]$ (Hsieh and Lavori, 2000), where R^2 is the variance in CCI explained by covariates. With alpha 0.05, power 0.90 and assumed R^2 0.50:

HR per SD of CCI	Required terminations (events)
0.70	165
0.75	254
0.80	422

Feasibility rule: the number of eligible polity terminations is counted from Seshat before Stage 1 submission. If fewer than 254 events are available, the minimum detectable effect is reported, Study 1 is designated as estimating rather than testing H1, and confirmatory weight shifts to Studies 2 and 3.

Feasibility audit (16 October 2026). The rule above was applied to the openly released Seshat Equinox-era sample (replication archive of Beheim et al., 2021: 33 Natural Geographic Areas, 334 unique polities). Termination type was derived from the Seshat code 'relationship to preceding (quasi)polity' recorded for each polity's successor within the same Natural Geographic Area. Results: 175 successions coded as continuity (censored), 30 polities right-censored at the end of the record, 65 successions uncoded, 16 with disputed codes, and 48 clear non-continuous terminations (28 conquest by elite migration or vassalage, 12 cultural absorption, 6 population replacement, 2 internal overthrow or revolt). Including disputed codes gives an upper bound of 64 events; if uncoded successions follow the coded rate of 20 to 27 percent non-continuous, the expected total is about 61 to 81 events. Clear events by era are 9

before 500 BCE, 30 from 500 BCE to 1500 CE and 9 after 1500 CE. This is well below the 254 events required. With 64 events, the minimum detectable hazard ratio per SD of CCI is 0.56 at 90% power and 0.61 at 80% power, under the assumed covariate R-squared of 0.50. Three consequences are pre-registered. (a) In this sample, Study 1 is an estimation study, not a confirmatory test of H1, and H7 is likewise underpowered. (b) H4a cannot be tested from Seshat succession codes, which record cultural continuity rather than political collapse (only 2 internal events); internal collapse will instead be coded from CrisisDB (168 crises) and the current Seshat release, and the event count will be re-audited on that combined sample before Stage 1 submission. (c) Because many Seshat polity boundaries mark archaeological periods rather than political ends, a termination is counted only for non-continuous successions. For Study 2, the D-PLACE Ethnographic Atlas (1,291 societies) is available, with high gods coded for 775 societies (198 with moralizing high gods); the Berezkin motif data must be obtained from the Michalopoulos and Xue (2021) replication archive. The audit table (one row per polity, with successor and termination type) and the analysis script accompany this protocol as polity_terminations_audit.csv and feasibility.py.
Re-audit on the current Seshat release (16 October 2026). The count was repeated on the current public Seshat release (seshat-db.com API: 864 polities, 35 Natural Geographic Areas, 45 regions) combined with CrisisDB (169 crises, 61 of them ending a polity). Succession codes yield 101 non-continuous terminations (53 conquest by elite migration, vassalage or hostile takeover; 15 absorption; 24 population replacement; 9 secession), excluding 2 territorial consolidations. CrisisDB adds 61 polity-ending crises (35 with conquest present, 14 with collapse or fragmentation and no conquest, 10 with civil war or decline only, 2 unclassified), of which at least 11 coincide with linked succession events. The combined sample therefore contains between 101 and 151 terminations, with 23 strict internal-collapse events (33 including civil war or decline). This remains below 254: the minimum detectable hazard ratio per SD of CCI is 0.69 at 90% power for 151 events (0.63 for 101), and 0.45 for 33 internal events. Coverage, not history, is the binding constraint: only 352 of 864 polities have a coded relationship to their predecessor, of which 29.3 percent are non-continuous. If the remaining 467 successions that have a predecessor (512 uncoded, less about 45 that open a regional sequence) were coded at that rate, Study 1 would reach about 240 succession events, a detectable hazard ratio near 0.74 at 90% power; this projection is optimistic, because uncoded successions are disproportionately prehistoric periods. Spot checks also found predecessor links that are clearly wrong (for example, a Warring States polity linked to a twelfth-century Jin state), so predecessor links will be verified against polity dates before analysis. Amendment: the first work package of Study 1 is to code the missing successions and polity-ending crises under the termination-type protocol in Section 3, blind to narrative content, and to repeat this audit; confirmatory status for H1 is granted only if the coded sample reaches 254 events. The extraction script is provided as seshat_current_audit.js. The termination-type coding instrument is the 'Termination-Type Codebook' (version 0.2 for this pilot, version 0.3 for the full run; the file is now titled 'Termination-Type Codebook v0.3 (Study 1)'); its reliability pilot uses 50 successions (40 uncoded, stratified by era: 21 before 500 BCE, 16 from 500 BCE to 1500 CE, 3 after 1500 CE; plus 10 calibration items with legacy Seshat codes, 5 continuous and 5 non-continuous), drawn with seed 20261016 and held in the 'Termination-Type Pilot Coding Sheet', with the calibration key in a separate adjudicator-only file. Ten of the 50 pairs overlap by more than 25 years, so the codebook's 'not a succession' category is expected to be used. The NGA-sequence audit also showed that only 377 of the 864 current polities sit in an NGA sequence (383 succession pairs, 75 uncoded); the remaining 487 polities, mostly from newer regional expansions, need their predecessor identified before they can be coded, which is the

larger share of the Study 1 coding work.
Provisional LLM pilot (16 October 2026, exploratory, not reliability evidence). Before human coding, the 50 pilot items were coded twice by independent instances of a large language model, each blind to the calibration key, barred from Seshat and CrisisDB, and required to cite two sources per item. Agreement between the two passes was 46 of 50 on termination type (Krippendorff alpha 0.895 on full codes, 0.893 on collapsed categories); because both passes share one model and one brief, their errors are correlated and this agreement is not evidence of human reliability. Both passes matched the legacy Seshat code on 7 of 10 calibration items and departed from it on the same three, one of which (the Spanish conquest of Oaxaca coded as continuity) appears to be a legacy error. A check of six cited pages found four that fully supported the claim, one that did so partly and one that did not contain it, so every LLM citation must be verified. The pilot exposed three codebook defects, corrected in version 0.2: the Q3 rule conflicted with archaeological phases (20 of 21 rule violations), 'external' was undefined when the predecessor was itself fragmented, and pairs separated by a missing intermediate polity had no code. Coders also flagged date or pairing problems in more than half of the items, which confirms that sequence verification must precede outcome coding. Among the 40 previously uncoded items, the provisional codes give 14 non-continuous terminations (35.0 percent, 95 percent Wilson interval 22.1 to 50.5), 8 of them internal collapses, compared with 29.3 percent non-continuity and only 9 percent internal events in the legacy codes. Extrapolated to the 467 uncoded successions, this implies about 264 terminations in total (interval 204 to 337), close to the 254 required, and about 120 internal collapses; the extrapolation assumes that polities outside NGA sequences behave like those inside them and rests on unvalidated codes, so it guides the coding budget but does not alter the feasibility rule. These LLM codes are held in a separate adjudicator-only file and play no part in confirmatory analysis unless an LLM coder later meets the validation standard in Section 5.
Study 2: with about 950 societies and an effective sample reduced by autocorrelation, standardized effects of about 0.12 to 0.15 are detectable at power 0.90. Study 3: for d = 0.15 at alpha 0.05 and power 0.90, 934 participants per arm (1,868 pooled) are required, about 234 per site across eight sites, inflated by 15% for exclusions.

6.11 Robustness

A specification curve across all pre-registered combinations of controls, corrections, imputation models and CCI weightings, reporting the share of specifications with effects in the predicted direction.

7. Separating truth-to-human-life from spread

A tradition can persist for two reasons: because its content supports human social life, or because its carriers conquered or out-reproduced rivals. NSH predicts the first. The design separates them in five ways:

1. **Competing risks (H4a).** Conquest-driven persistence predicts an effect on conquest avoidance only. NSH predicts an effect on internal collapse, which military power does not prevent.
2. **Reverse transmission (H4b).** When conquerors adopt the narratives of the conquered, adoption cannot be explained by the military success of the source tradition.
3. **Conditioning on power.** All Study 1 models adjust for agricultural productivity and

military technology, the best-supported drivers of complexity (Turchin et al., 2022), and for population growth as a fertility proxy.

4. **Crisis recovery.** Recovery after a shock, measured in CrisisDB, depends on internal cohesion rather than conquest.
5. **Experiment (H6).** Random assignment removes conquest and fertility entirely, testing whether content itself changes cooperation with strangers.

8. Interpretation matrix

Pattern of results	Interpretation
H1 to H7 supported	Narrative content is a selected trait that supports societal endurance through cooperation beyond kin, independently of material power, its vessel and its era
H1 and H2 supported, H4a fails	Content predicts endurance only through conquest; traditions persist because carriers win, not because content serves social life
H1 supported, H3 fails (reverse lag dominant)	Content is a product of successful societies rather than a cause
H1 supported, H5 fails	The effect belongs to supernatural belief, not to narrative content
H6 supported, H1 fails	Content changes individual behaviour but this does not scale to polity endurance
H7 fails	Content effects are era-specific rather than atemporal
Equivalence tests succeed for H1 and H6	NSH is falsified

9. Threats to validity and mitigations

- **Text survival bias.** Elite and written traditions survive more often. Mitigation: writing and literacy as covariates; oral folklore in Study 2; missingness modelling rather than exclusion.
- **Coder knowledge of history.** Mitigation: redaction, blinding check, coders from unrelated specialisms.
- **Reverse causation.** Mitigation: lagged and cross-lagged designs (H3) and the experiment (H6).
- **Construct validity of endurance.** Long-lived polities are not necessarily good places to live. Mitigation: crisis recovery and impersonal trust as additional outcomes; endurance is interpreted as persistence of social life, not as welfare.
- **Measurement invariance.** Moral domains may mean different things across cultures. Mitigation: invariance testing of the CCI; results reported per macro-region.
- **Ecological inference.** Polity-level associations do not establish individual mechanisms. Mitigation: Study 3.
- **Triangulation.** Following Munafò and Davey Smith (2018), conclusions rest on

convergence across designs with different biases, not on any single study.

10. Boundary of the empirical claim

If the hypotheses are supported, the study establishes that the moral content of human narrative traditions is non-arbitrary: it is a selected trait, it predicts the endurance and recovery of societies independently of material power, it causally increases cooperation with strangers, and its effect is stable across eras. In the terms of the author's philosophical framework, this would empirically confirm the content of meta-truths, the narratives that crossed the Weismann barrier and survived selection against the environment of human life together.
No result of this study can establish that sustaining human social life is good. That premise is normative, and Hume's distinction between what is and what ought to be places it outside empirical test. Any attempt to justify it ends in one of the three outcomes of the Munchhausen trilemma: infinite regress, circularity, or a chosen starting point. The study therefore states this premise as an axiom rather than a finding. Science can confirm what meta-truths contain and what they do; the choice to live by them remains a commitment that no data can make on anyone's behalf.

The same boundary governs moral objectivity. If H9a holds, cooperative moral content is objective in a precise, limited sense: stance-independent facts about which norms solve recurrent problems of human cooperation, discovered repeatedly by groups that did not learn them from one another. It would not establish mind-independent moral realism, because convergence is equally explained if selection favours norms that serve fitness rather than truth (Street, 2006; Joyce, 2006). Within the framework tested here that is not a defeat: an evolutionary truth is defined as what is true of humans, not of atoms.

Author's note (philosophical interpretation; not part of the empirical claims or inference criteria of this protocol).

Asking whether sustaining human society is good already subjects reality to a human question. It is the same anthropomorphization as calling genes selfish. Reality gives no answer because it has none to give. Humanity's genetic evolution, and the eternal unconscious it built, is directed toward nothing: selection is blind to any goal, drift is pure chance, and both unfold only relative to the replicator's environment on a rock in the cosmos. From this point of view there is no eternal reason, and the leap must be taken absurdly.

But humanity's eternal conscious points somewhere. Across societies that never met and eras that never touched, humans have discovered and rediscovered the same unchanging ideal: the convergence of all good, that which selects among men in the eternal hierarchy of men. That is God. The convergence is not only a matter of faith: in the ethnographic records of 60 societies, Curry, Mullins and Whitehouse (2019) found the same seven forms of cooperation judged morally good almost without exception wherever they appeared, and H7 of this protocol tests whether their social effects are equally stable across eras. What humans converged on, however, is the content, not the name: only a quarter of the coded societies in the Ethnographic Atlas believe in a moralizing high god. God, in this framework, is the name for that convergence rather than one of its independent discoveries, and H9 tests the convergence itself. If it holds, it makes morality objective in the one sense evolution allows: true of us, whatever we believe

about its source.

Hierarchies offer a second test. If what selects among men is the convergence of good, then hierarchies led by those who serve should outlast those held by force. Peterson (2013), following de Waal, observes that even among chimpanzees rule by strength alone is unstable, and that the best player is the one others keep inviting to play. The evidence so far is mixed: a tolerant baboon culture outlasted the departure of every male present at its start, yet its alphas held power no longer than elsewhere; at Gombe the most forceful alpha held power longest; and across thirty premodern states, good government did not significantly lengthen a state's life. H8 tests the claim; this note does not assume it. So the leap is absurd from the objective view, yet it is not a leap into nothing. It is a leap toward what every tradition has found.

The value of human life together is the starting point, an axiom no system can prove from within. Gödel's second incompleteness theorem offers a formal parallel: a consistent formal system rich enough for arithmetic cannot prove its own consistency from within itself. Science stands on the same kind of axiom, the faith that order can be made from chaos. So the honest conclusion is not that God is proven. It is that science can confirm the content of meta-truth, whether the stories humanity has kept are adaptive rather than arbitrary, while living by them remains the leap. The proof ends where the leap begins.

11. Procedures

- **Data and code availability.** All coded data, codebooks, prompts, model versions and analysis scripts deposited openly at Stage 2.
- **Deviations.** Any deviation from this protocol is logged with date and reason and reported in a dedicated section.
- **Exploratory analyses.** Clearly labelled as exploratory and excluded from confirmatory claims.
- **Ethics.** Study 3 requires institutional ethics approval at each site, informed consent and incentive payment in local currency.

12. Deviations from protocol

Logged 17 October 2026, before any association between narrative content, ruler accountability or moralizing gods and polity outcomes had been computed. The protocol as it stood before these deviations is frozen in Google Drive as 'FROZEN 2026-10-17 Narrative Selection and Societal Endurance: Registered Report Protocol (pre-analysis snapshot)'. Each entry states what changed, why, and what the change does to the strength of inference.

D1. No Stage 1 review and no public registration. The protocol has not been peer reviewed or registered. The frozen Drive copy fixes the plan, but it is not a public time-stamped registration that a third party can verify. No result reported here can be described as a Registered Report finding. Remedy open to the author: deposit the frozen copy on OSF Registries; the deposit then precedes any analysis not yet run.

D2. Data access before freezing. The header states that the plan was fixed before data access. The feasibility audits in Section 6.10 read Seshat polity dates, Seshat succession codes and CrisisDB, which contain the outcome, as well as the D-PLACE and Seshat distributions cited in Section 1.2. No predictor was linked to any outcome in those audits.

D3. Coders. Section 5 requires two trained human coders and admits a language model coder only after it matches human agreement on at least 500 held-out items. No human coders are available. All new coding (termination type and any other item-level judgement) is therefore done by independent instances of a large language model working from the frozen codebook, with a separate instance as adjudicator, and no human validation set exists. Agreement between instances measures the consistency of one model, not human reliability, and errors the instances share are invisible to it. Every analysis that uses these codes is exploratory, and the condition in Section 5.3 is not met in this run.

D4. Narrative content measure. A language model cannot be blinded to the narratives of known polities: it recognizes canonical texts and knows how the polities that carried them ended, so model-judged CCI codes would breach the blinding rule in Section 5.2. The CCI is instead scored by a fixed dictionary, the Morality-as-Cooperation Dictionary (MAC-D; Alfano, Cheong and Curry, 2024), applied to public-domain translations of each tradition's canonical texts and linked to polities through the Seshat religion variables. Dictionary scoring involves no coder judgement and is outcome-blind by construction. Its costs: word frequencies measure how salient a moral domain is, not whether it is endorsed; translation choices change the counts; the moral scope weight cannot be scored by dictionary and is dropped; and polities that share a tradition share a score, so the effective number of independent CCI values is the number of traditions, not the number of polities.

D5. Study 3 not run. A randomized experiment needs real participants, ethics approval and incentive payments. Simulated participants would be fabricated data. H6 remains untested, and the rows of the interpretation matrix that depend on H6 cannot be evaluated.

D6. Later deviations are appended to this list with their dates.

D7. Termination coding unit and codebook v0.3 (17 October 2026). The coding item became the polity rather than the predecessor-successor pair, because the outcome is how each polity ended. Codebook v0.3 is v0.2 with the coder naming the successor (category S2 and code NA removed) and an end-year estimate added. The pilot items were recoded fresh under v0.3 and kept in the sample; a sensitivity analysis excludes the 35 in-frame polities that appeared in the pilot.

D8. Coding scope and double coding. Only the 429 polities that enter at least one model were coded (those with a mappable narrative tradition, or a known Seshat value for moralizing enforcement of rulers or for moralizing supernatural punishment), not all 864. Each was coded once; a random 20 percent (86 items, seed 20261017) was coded again by independent instances, and the 7 disagreements on termination type were adjudicated by a third instance. Section 5 requires two coders on every item.

D9. Covariates. Most Seshat values are not dated within a polity, so covariates are polity-level averages rather than century-level time-varying values. Agricultural productivity and kinship intensity are not in the public Seshat API and are omitted from H2. Social complexity is the first principal component of nine characteristics built from current Seshat variables (67.5 percent of variance on average), and military technology is the summed within-category share of 45 warfare variables in six categories; both simplify Turchin et al. (2018, 2022). Missing covariates were imputed 50 times by chained equations (Bayesian ridge with posterior draws), with the event indicator, the Nelson-Aalen cumulative hazard and macro-region in the imputation model.

D10. Models. Shared frailty by NGA and the spatial covariance term could not be fitted with the software available (no R). They are replaced by standard errors clustered on narrative tradition, a tradition-level permutation test for the CCI as the primary inference, and sensitivity analyses with macro-region strata and region clustering. Fine-Gray models were fitted with custom code checked against statsmodels and by simulation. Two model fits did not converge (one

imputation in each of two H4a models) and were dropped from pooling. After the analysis script was frozen (SHA-256 beginning a2fcd350), the only changes were for numerical robustness; the diff is archived with the code.

D11. Hypotheses not tested in this run: H3 (the dictionary CCI is constant within a tradition, so it cannot be lagged within polities); H4b (the conquest-episode coding was not done); H6 (Study 3, see D5); H9a (it needs new coding of ethnographies, and eHRAF is not accessible here); the narrative Ruler Constraint item of H8 (a language model cannot code it blind); and Study 2 (the Berezkin motif data were not obtained in this run).

D12. H9b. The D statistic was computed on a global tree assembled from the D-PLACE Glottolog classification trees (topology only, unit branch lengths, families joined at a root, polytomies resolved at random), with traits collapsed to one value per language. No cooperative-norm codes exist for the same societies, so the comparison in H9b's falsification condition cannot be made; only the signal of deity belief is estimated.

D13. Termination roles and codebook sensitivity analyses (17 October 2026). In the primary models secession (T5, one polity in the H1 sample) was grouped with absorption and replacement, rather than counted as internal collapse as the codebook's analysis-role column specifies, and insufficient-evidence codes (T9) were handled through Q3 (event if Y, censored if N, dropped if U) rather than multiply imputed; no T9 item entered the H1 sample. The codebook's pre-specified sensitivity analyses were run afterwards with 20 imputations and are reported under H4a in 13.3.

D14. Timing of the addendum (22 October 2026). The Section 14 plans were written after the Study 1 results in Section 13 were known, though before the data each plan uses were examined. The person fixing them knew that the Study 1 content effects were null. This does not bias the new tests toward any particular result, but they are not independent of the first run in the way a preregistered confirmatory study would be.

D15. Dictionary file. The MAC-D files used come from the eMACDscore package (github.com/medianeuroscience/eMACDscore). The vice file matches the size of the OSF original (osf.io/86qry). The virtue file is 28,332 bytes, 9 bytes smaller than the OSF listing. The OSF file could not be downloaded from this environment, so the difference (probably one short entry) is unidentified. Every CCI score in Sections 13 and 15 uses the package file.

D16. Study 2 smallest effect of interest. The bound of 0.10 SD per SD of CCI (Section 14.1) was set without calibration. The positive control, the published trickster-punishment measure, is about 0.03 SD per SD in the immigrant sample. So an equivalence result at 0.10 SD rules out only effects about three times the size of the strongest known folklore effect. In Section 15, any immigrant effect below 0.10 SD is reported as small, not as absent.

D17. Study 2 measurement. Motif descriptions are one or two sentences long, and only 26 percent of motifs contain a MAC-D virtue term. Deference terms appear in 0.35 percent of motifs and fairness terms in 0.08 percent, so the z-scores of these two domains rest on a handful of motifs, and the folklore CCI mostly reflects family and heroism words. The index was kept as specified.

D18. H8-3 covariate. The plan adjusted crisis severity and ruler removal for population and era, but the first script (h8_improved.py) included only era. A corrected model with log population (h8_3_population.py, complete cases) is reported alongside it in Section 15.

D19. H4b detail fixed in code. Section 14.2 did not specify four rules: which tradition counts when coders correct the database; how unknown conqueror writing is handled; the minimum number of events; and the number of permutation draws. These were fixed in the header of h4b_analysis.py before the adjudicated codes were read. Coders worked under a shared web-search budget of 8 searches per batch of 10 episodes.

D20. H9a sample and texts (23 October 2026). Full public-domain English ethnographies published before 1929 are unevenly available. The 46 verified candidates covered 36 top-level families: 21 were in the Americas, and none was found for Uralic, Turkic, Tungusic or Caucasian peoples. The fixed rule took one society per family at random, then up to 6 per region with a random fill. It gave 30 societies: 13 in the Americas, 6 in Asia, 5 in Oceania, 4 in Africa and 2 in Europe and northern Eurasia. Several texts are topical rather than general accounts: a calendar history (Kiowa), a mythology (Tsimshian), a study of religion (Khoikhoi), a study of magic (Orokaiva), a language-and-folklore volume (Nandi) and a multi-tribe survey (Natchez). Two are not fieldwork monographs: a missionary's account (Mongols) and a social study (Russian peasants). One predates academic ethnography (Lane on Cairo, 1836). The Shan were judged distinct from the excluded Central Thai. Monographs bound in larger volumes were cut at their headings, with no regard to content; the Mailu paper, for example, was taken from its journal volume. Passages were extracted by the fixed rule: 2,368 items, with up to 12 per behaviour per society, and fewer where a text had fewer matching paragraphs (fairness passages were scarce).

D21. H8 follow-up (23 October 2026). The tests of whether ruler removal and internal collapse are costly to the society were added on the author's request, after the H8 results were known. They are not in Section 14. Their plan was fixed in h8_followup_plan.md before the outcomes were computed. Two checks were added after seeing the results and are labelled as such: the survivor-only model, and complexity built only from non-scale measures. The earlier suggestion in this run that deposition might reflect accountability is withdrawn, because the follow-up does not support it.

D22. Errors found by the independent verification (23 October 2026). First, the MAC-D scorer keyed dictionary entries on whitespace only. Because text tokens never contain hyphens, the 39 hyphenated entries (for example 'father-in-law', 'give-and-take', 'even-handed') could never match. Corrected, the tradition CCIs correlate 0.9999 with the originals (largest change 0.009). Greek, Norse-Germanic and Mesopotamian, which were nearly tied, swap ranks 4 to 6; the Study 1 results are unaffected in substance. In Study 2 the bug changes 4 motif flags, and the political-scale estimate moves from +0.077 to between +0.071 and +0.076. Second, Study 2 dropped 8 motifs whose incidence columns end in an underscore (effect below 0.001). Third, the first Study 2 summary reported the immigrant model without the individual controls the plan specified. The planned model gives +0.015 SD (90 percent interval -0.013 to 0.043) for the CCI, +0.036 ($p = 0.002$) for the net index, and $b = 1.07$ ($p = 0.008$) for the trickster control; Section 15 reports these. The stemming of -eth and -est endings, which the verifier flagged, is specified in the corpus rules and was kept.

D23. Revised H8 test (23 October 2026, at the author's request). Seshat's moralizing enforcement of rulers measures what supporters claimed about rulers, not what rulers did. H8 is therefore re-tested on conduct, using expert-coded V-Dem data for autocracies from 1789 to 2025 (Section 14.5). The plan was fixed in h8r_plan.md before the models were run. After freezing, two bugs were fixed (h8r_changes_after_freeze.diff). Columns named C and D clashed with the formula syntax and were renamed. Models were refitted on complete cases so that absent fixed-effect levels do not produce empty columns and undefined standard errors. Neither fix changes any estimate. The original Seshat-based H8 results stay in Sections 13 and 15 as the test of the original exposure.

13. Results of the exploratory run (17 October 2026)

Everything in this section is exploratory under Deviations D1 to D13. No test survived the Holm-Bonferroni correction, and no result here is confirmatory. Code, data extracts and logs are kept with the analysis files (polities_frame.csv, final_codes.csv, cci_traditions.csv, covariates.csv, analysis.py, results.json, h9b_results.json).

13.1 Coding frame and termination codes

Of 864 polities in the current Seshat release, 430 enter at least one model; 429 were coded and 1 is right-censored at the present. Termination types across the 429: continuity 164, conquest 163, internal collapse 67 (fragmentation or civil war 30, abandonment 2, revolution or coup 35), absorption 11, population replacement 2, secession 2, voluntary union 2, insufficient evidence 18. The sample therefore holds 245 terminations of known type, close to the 254 the power analysis required; but only 176 of them fall in polities with a scorable narrative tradition. Agreement on the 86 double-coded items: termination type 79 of 86 (Krippendorff alpha 0.89 on full codes, 0.92 on collapsed categories, 0.90 on termination yes or no); succession category alpha 0.91; end of central authority 0.96; prior internal crisis 0.80; confidence 0.47. Both coders are instances of one model, so these figures measure its consistency, not human reliability (D3). An audit of 20 random items found that the cited pages fully supported 16 codes and partly supported 4, with none contradicted; 5 had date discrepancies of 1 to 30 years between code and source.

13.2 Cooperative Content Index

CCI by tradition (mean z-score across the seven MAC-D virtue domains), from lowest to highest: Daoism -0.39, Buddhism -0.38, Judaism -0.23, Christianity -0.21, Zoroastrianism -0.20, Egyptian -0.18, Hinduism -0.07, Confucianism -0.05, Roman 0.07, Greek 0.09, Norse-Germanic 0.09, Mesopotamian 0.09, Jainism 0.25, Islam 0.41, Shinto 0.69. Corpora range from 21,708 tokens (Mesopotamian) to 790,046 (Christianity). The ranking shows the measure's main weakness: Shinto scores highest mainly because the Kojiki's genealogies are dense with kinship words (113 Family-domain terms per 10,000 tokens, about twice any other tradition), which is salience, not endorsement of family obligations (D4). Of the 430 framed polities, 261 have a tradition under the primary rule; 253 enter the H1 sample, falling into 19 distinct tradition combinations from 15 traditions.

13.3 Study 1: narrative content and endurance

H1 (content predicts endurance): hazard ratio per SD of CCI 1.08 (95 percent CI 0.90 to 1.28; tradition-level permutation $p = 0.51$; 253 polities, 176 terminations). Not supported. Equivalence is not established either, because the 90 percent CI (0.93 to 1.25) extends past the 1.11 bound; the data exclude a protective effect stronger than about 0.90 per SD but not a small effect in either direction.
H2 (robust to drivers of power): adjusted hazard ratio 1.08 (0.92 to 1.25; permutation $p = 0.47$). Not supported. Of the controls, only iron weapons reached $p < 0.05$ (hazard ratio 2.77, 1.03 to 7.42).
H4a (internal versus external exit): Fine-Gray sub-hazard ratio for internal collapse 1.15 (0.80 to 1.66; permutation $p = 0.28$; 49 events) and for conquest 1.04 (0.78 to 1.38; 117 events); cause-specific estimates 1.18 and 1.05. Not supported: CCI does not lower the internal-collapse hazard. Codebook sensitivity analyses (D13) leave this unchanged: counting secession as

internal collapse gives 1.19 (0.85 to 1.67); merging absorption with conquest gives 1.16 for internal collapse and 1.01 for conquest; and reclassifying conquests preceded by an internal crisis (79 polities) as internal collapse gives 1.18 (1.00 to 1.38), again in the direction opposite to the prediction. The sample contains no population-replacement endings.
H5 (vessel independence): with moralizing supernatural punishment in the model the CCI hazard ratio is 1.07 (0.93 to 1.24), unchanged, but there is no CCI effect for the vessel to absorb. The moralizing punishment coefficient itself is uninformative (hazard ratio 8.0, 0.8 to 80), because 85 percent of coded polities have it.
H7 (temporal invariance): era-specific hazard ratios (adjusted) are 0.58 before 500 BCE (34 polities, 18 terminations), 1.03 from 500 BCE to 1500 CE and 1.09 after 1500 CE; interaction Wald p = 0.085 adjusted (0.007 unadjusted; Holm-adjusted 0.43). The early-era estimate rests almost entirely on two traditions (Mesopotamian 13 polities, Egyptian 12), and the Wald test ignores the covariance between the two interaction terms, so this is not evidence for or against temporal invariance.
Robustness. The proportional-hazards assumption holds (Schoenfeld rho 0.05, p = 0.55). Dropping one tradition at a time gives hazard ratios from 0.96 (without Islam) to 1.14 (without Buddhism). Other sensitivity analyses give 0.95 to 1.16: complete cases 1.05 (n = 96), the virtue-minus-vice index 0.99, coder end dates 1.06, macro-region strata 1.07, excluding pilot polities 1.04, and assigning traditions by official religion 1.16 (1.01 to 1.33). Across the 72-specification curve the median hazard ratio is 1.07; 15 percent of specifications fall below 1, none significantly, while 7 of 72 (all using official-religion assignment) are significant in the direction opposite to the prediction. A separate script written by an independent instance, using statsmodels, reproduced the H1 sample (253 polities, 176 terminations, 49 internal collapses, 117 conquests) and the H1 hazard ratio (1.0755) exactly.

13.4 H8: rulers under moral sanction

Among 363 polities with a known Seshat code for moralizing enforcement of rulers (294 present, 69 absent; 60 internal collapses), the cause-specific hazard ratio for internal collapse is 3.04 (1.18 to 7.85) and the Fine-Gray sub-hazard ratio 2.20 (0.82 to 5.93); after adjustment for the H2 covariates it is 1.32 (0.34 to 5.12). The conquest sub-hazard is also higher (1.89, 1.11 to 3.24). H8a is not supported: the unadjusted association runs opposite to the prediction and vanishes with adjustment, which suggests that the code tracks scale and era rather than ruler conduct. H8b: mean crisis severity at a polity-ending crisis is 0.81 where rulers faced moralizing enforcement (n = 43) and 1.00 where they did not (n = 6; Mann-Whitney p = 0.50), too few cases to test. The Seshat variable measures supernatural sanction on rulers, not service-based leadership, so H8's core claim remains untested.

13.5 H9b: belief in a moralizing high god

Across 692 languages (775 Ethnographic Atlas societies), a moralizing high god (EA034 = 4) is present in 23.6 percent. Its phylogenetic signal is strong: D = -0.20 ($p < 0.001$ against a random distribution; $p = 0.92$ against a Brownian threshold model), meaning it is at least as clustered by descent as a trait evolving by Brownian motion; across 10 random resolutions of the tree's polytomies D ranged from -0.23 to -0.16. A high god of any kind (64.0 percent) is less clustered (D = 0.19; range 0.16 to 0.23) but still far from random. The odds of a moralizing high god rise 2.37-fold per level of jurisdictional hierarchy above the local community (95 percent CI 1.86 to 3.02), from 6.3 percent in societies with no such level to 61.1 percent in those with four. Belief in

a moralizing God therefore travels with descent and with political scale, as expected of a trait spread by inheritance and contact rather than discovered independently. Because H9a was not run, the convergence of cooperative moral content itself was not tested here.

13.6 Interpretation

In this run, the moral content of canonical texts, as scored by a fixed dictionary, does not predict how long polities lasted, how they ended, or whether they collapsed from within. The estimates rule out a moderate or large protective effect but cannot exclude a small one. This is a weak test of the narrative selection hypothesis, for five reasons: the effective sample is 15 traditions, not 253 polities; the dictionary measures how often moral domains are mentioned, not whether a text endorses them; polities are assigned the canon of their dominant religion, which ignores local and oral narratives; outcomes were coded by one model without human validation; and two thirds of terminations are conquests, which the hypothesis does not claim content prevents. The result most robust to these limits is H9b: belief in a moralizing God tracks descent and the scale of political organization. A confirmatory test needs human coders validating at least a subset of termination codes, an endorsement-based content measure coded blind on redacted passages, far more traditions including oral ones (Study 2), and Stage 1 review before any of it is run.

14. Addendum: analysis plans for H4b, H8, H9a and Study 2 (fixed 22 October 2026, before the data they use were examined)

These plans were written after OSF reopened and the Michalopoulos and Xue (2021) replication archive was obtained, but before any folklore score, adoption code, ethnographic valence code or new H8 model had been computed. They inherit Deviations D1 to D13: coders are instances of one language model, and every result is exploratory. A copy of the protocol as it stood when this addendum was added is frozen in Drive as 'FROZEN 2026-10-22 Narrative Selection and Societal Endurance: Registered Report Protocol (addendum, pre-analysis)'.

14.1 Study 2: folklore content and trust

Data: the replication archive of Michalopoulos and Xue (2021; Harvard Dataverse, doi:10.7910/DVN/IXOHKB): Berezkin's 2,564 motifs with his own English titles and descriptions, motif incidence across 1,245 Ethnographic Atlas groups and 199 countries, and their outcome files (country trust, European Social Survey second-generation immigrants, Ethnographic Atlas variables).
Measure: each motif's English title and description are scored with MAC-D under the rules of the corpus specification (Section 3 of corpus_spec.md). A motif carries domain d if it contains at least one virtue term of d. For each group or country, the domain score is the share of its motifs carrying d (incidence-weighted for countries). The folklore CCI is the mean of the seven z-scored domain shares, with z computed across the units of each dataset; the net index uses virtue minus vice flags. Units with fewer than 5 motifs are excluded.
Models: (S2-1) country trust, ln(average WVS trust) on CCI with the documentation controls of Michalopoulos and Xue (log first publication year, log number of sources), robust errors, without

and with continent dummies; secondary outcome GPS trust. (S2-2) second-generation immigrants in the European Social Survey: ln(trust) on the CCI of the parents' origin country with their individual controls and host-country-by-round fixed effects, errors clustered by origin country, on their exact sample. (S2-3) political scale in the Ethnographic Atlas: levels of jurisdictional hierarchy above the local community (v33) on CCI with the documentation controls and continent fixed effects, errors clustered by language family, groups with more than 4 motifs; then adding high-god dummies (v34) as the Study 2 analogue of H5. Positive control: the authors' trickster-punishment measure is re-estimated in S2-1 and S2-2 with the same code; if it does not reproduce their published sign and approximate size, the pipeline is treated as faulty and the CCI results are not interpreted. Prediction: positive CCI coefficients. Smallest effect of interest: 0.10 SD of the outcome per SD of CCI; equivalence by two one-sided tests at 90 percent.

14.2 H4b: reverse transmission

Frame: Study 1 polities coded as conquered (T2) whose narrative tradition has a CCI score (117 episodes). Each episode is coded by independent model instances, blind to CCI values: the conquerors' tradition at the time of conquest (one of the 15 scored traditions, another named tradition, or unknown); whether, within 100 years, the conquering ruling group adopted the conquered population's tradition as its own (yes, partial, no, unknown); and whether the conquering elite used writing at the time of conquest. Episodes where conquerors already shared the conquered tradition are excluded, since no adoption is possible. A random 20 percent is coded twice and disagreements are adjudicated.
Analysis: logistic regression of adoption (yes = 1, no = 0; partial excluded, and counted as yes in a sensitivity analysis) on the conquered tradition's CCI (z-scored in the H4b sample), controlling for the conquered polity's social complexity (prestige) and writing, and the conquerors' writing; errors clustered by conquered tradition, with a tradition-level permutation test as the primary inference. Secondary: where both traditions are scored, adoption on the CCI difference (conquered minus conqueror), the two-alternative form of conditional logit. Prediction: odds ratio above 1. Falsified if the odds ratio is below 1 or its 90 percent interval lies within 0.80 to 1.25.

14.3 H9a: moral valence of the seven cooperative behaviours in a new sample

Sample: one society per top-level Glottolog language family, excluding the 60 Probability Sample File societies used by Curry, Mullins and Whitehouse (2019), drawn from societies with a full English-language ethnography published before 1929 and available as public-domain full text. Target 30 societies. The candidate list and the final sample are fixed and recorded before any passage is coded.
Passages: each ethnography is cleaned as in the corpus specification and split into paragraphs. For each MAC-D domain, paragraphs containing at least one virtue or vice term of that domain are candidates, and up to 12 per domain per society are drawn at random (seed 20261022). Using the dictionary's own term lists, including its vice terms, avoids hand-picked keywords and lets disapproval surface.
Coding: independent model instances, not told the hypothesis, code each paragraph: whether it describes members of the society performing, expecting or evaluating the domain's behaviour

(helping kin, loyalty to the group, reciprocity, bravery, deference to superiors, fair division, respect for property); and if so, the society's valence toward it (positive: approved, praised, obligatory; negative: condemned, punished, shameful; none stated), with the evaluative words quoted. A random 20 percent is coded twice. Criterion (Section 2): H9a is falsified for a behaviour if it is judged bad in more than 5 percent of the instances in which its valence is coded; each percentage is reported with a 95 percent Wilson interval.

14.4 H8: stronger tests with existing data

(H8-1) Internal-collapse hazard by moralizing enforcement of rulers within Natural Geographic Areas: cause-specific Cox model stratified by home NGA, so that polities are compared only with others in the same region, adjusted for social complexity, population and era. (H8-2) The same model restricted to literate states (writing present). (H8-3) Crisis severity across all 169 CrisisDB crises, not only polity-ending ones: count of civil war, extermination and collapse present, by moralizing enforcement, with errors clustered by polity, adjusted for population and era; and the probability that a crisis ends with the ruler deposed or assassinated. Predictions: hazard ratio below 1 (H8-1, H8-2) and lower severity and deposition (H8-3). The narrative Ruler Constraint item and an independent measure of accountable rule (for example the Blanton and Fargher codes for 30 premodern states, which are not public) remain unavailable.

14.5 H8 revised: ruler conduct and long-term outcomes (fixed 23 October 2026, before the models were run)

Exposure: what rulers do, measured in V-Dem v15 (expert-coded, 1789 to 2025) for autocratic country-years. Dominance is the mean of political killings, torture and civil society repression. Cooperation is the mean of executive respect for the constitution, judicial and legislative constraints, and public-goods spending. Strategy is cooperation minus dominance, standardized. (A) Religious legitimation (1900 to 2023): association with religious freedom, repression, constraints, public goods and personalist legitimation, with region or country fixed effects, compared with nationalist and socialist legitimation. (B1) Forward outcomes from each autocracy-year: growth of GDP per head over 10 and 20 years, change in life expectancy over 20 years, civil war onset and coups within 10 years. Controls: income, population, decade and region, with errors clustered by country. (B2) For each autocratic regime: whether it ended violently from within (coup, assassination, civil war defeat, uprising) rather than in an orderly way. (B3) Regime duration. Supported if strategy predicts better outcomes in all of B1 and fewer violent ends in B2. The full plan is in h8r_plan.md.

15. Results of the addendum analyses (22 to 23 October 2026)

All results below are exploratory (Deviations D1 to D23). Study 2, H8 and the revised H8 use no AI-made codes. H4b and H9a rest on codes made by independent instances of one AI model, with no human validation. The per-hypothesis report 'Narrative Selection and Societal Endurance: Results by Hypothesis' gives reasons and better tests for each result.

15.1 Study 2: folklore content and trust

Positive control. The authors' trickster-punishment measure reproduced their published findings. Across countries, b = 1.86 ($p < 0.001$, 0.34 SD per SD, 104 countries); among second-generation immigrants, b = 1.07 ($p = 0.008$, about 0.03 SD per SD). The pipeline is therefore treated as sound. An independent re-derivation matched every estimate either within rounding or within small differences from tokenization and model specification (Deviation D22).
Cooperative content (CCI). Country trust: -0.02 SD per SD (90 percent interval -0.24 to 0.19). GPS trust: +0.11 (-0.16 to 0.39, 76 countries). Second-generation immigrants, with the authors' individual controls: +0.015 SD (90 percent interval -0.013 to 0.043; 6,558 people, 117 origin countries). This is equivalent to zero within the 0.10 SD bound, but that bound is miscalibrated (D16). The net index (virtue minus vice) gives +0.036 SD among immigrants ($p = 0.002$), the same size as the trickster effect. Political scale in the Ethnographic Atlas: +0.077 SD per SD ($p = 0.18$; 1,105 groups, 71 families), and +0.063 ($p = 0.17$) with high-god belief controlled. Prediction (positive CCI coefficients) not supported for the CCI. There is a small positive signal for the net index. Correction to Section 14.1: the archive covers 1,265 Ethnographic Atlas groups, not 1,245.

15.2 H8 with the original exposure (Seshat moralizing enforcement of rulers)

Internal-collapse hazard by moralizing enforcement of rulers:

- within Natural Geographic Areas: HR 4.50 (0.50 to 40.6; 261 polities, 46 internal collapses, 33 NGAs);
- literate states: HR 1.69 (0.51 to 5.60; 269 polities, 50 internal collapses);
- literate states within NGAs: HR 3.01 (0.49 to 18.7).

Across 135 CrisisDB crises with consequences coded, severity was not lower where rulers faced moral sanction (0.74 versus 0.54, $p = 0.19$; +0.12, $p = 0.51$ with population). Rulers were more often deposed or assassinated (67 versus 22 percent; odds ratio 7.4, 95 percent interval 2.6 to 21; 5.5, $p = 0.006$, with population). Not supported.
Follow-up on whether removal and internal collapse are costly (D21). Removal crises more often involved civil war (66 versus 35 percent, Holm $p = 0.003$). New public goods followed none of the 93 removals, against 5 of 60 other crises (Fisher $p = 0.008$). After a removal crisis the polity ended non-continuously sooner (HR 1.56, 1.04 to 2.35), and 73 percent of removal crises came within 10 years of the polity's end. Compared with continuity in the same region, internal collapse was followed by a less complex successor: -0.15 (-0.28 to -0.01) on the mean of nine z-scored measures, and -0.11 (-0.22 to -0.01) with institutional measures only. It was also followed by a smaller successor territory and an interregnum 19 years longer. Conquest showed no such loss. Moral sanction of rulers did not soften either cost.

15.3 H8 revised: ruler conduct (V-Dem, Section 14.5)

Sample: 21,576 autocratic country-years in 202 countries, 1789 to 2025. The strategy score (cooperation minus dominance) is standardized. Effects are per SD, with errors clustered by country.
Part A, religious legitimation, 1900 to 2023. Effects are for moving from no religious legitimation

to full, between regions and then within countries over time:

- freedom of religion: -2.0 SD (-2.8 to -1.2); within countries -1.0 (-1.6 to -0.4);
- political killings: +0.95 (0.41 to 1.49); within +1.29 (0.74 to 1.84);
- torture: +1.01 (0.48 to 1.54); within +1.05 (0.12 to 1.99);
- judicial constraints: -1.11 (-1.69 to -0.52); within -0.93 (-1.57 to -0.28);
- public goods: -1.14 (-1.93 to -0.34); within -1.17 (-2.49 to 0.15);
- portrayal of the leader as extraordinary: +1.10 (0.53 to 1.67);
- respect for the constitution and legislative constraints: no clear difference.

Nationalist and socialist legitimation share parts of this profile, but only religious legitimation combines all of them. The constraint reading is not supported; the tool-of-power reading is.
Part B1, forward outcomes:

- growth of GDP per head over 10 years: +2.0 percent (0.6 to 3.5); over 20 years: +3.1 percent (0.2 to 6.0);
- life expectancy over 20 years: +0.52 years (0.17 to 0.87);
- civil war onset within 10 years: +0.7 percentage points (-1.7 to 3.0);
- coup within 10 years: +1.7 points (-1.8 to 5.1).

With country fixed effects the growth and health estimates shrink and are not significant. Within countries, coups became more likely as autocracies became more cooperative: +8.8 points (4.1 to 13.4).
Part B2, how regimes end. Among 1,099 autocratic regimes ending violently from within or in an orderly way, the odds ratio for a violent internal end was 0.75 per SD (0.63 to 0.89; 63 percent in the least cooperative quarter versus 39 percent in the most cooperative). For defeat in war or foreign intervention it was 0.51 (0.41 to 0.63).
Part B3, duration: HR 0.97 (0.92 to 1.02).
Verdict: partly supported. The ending test, growth and life expectancy moved as predicted; civil war and coups did not; and no outcome moved significantly the wrong way. The pre-set rule required all of them, so the strict verdict is inconclusive. As the model in Section 2 predicted, cooperative and dominance regimes lasted equally long but ended differently.

15.4 H4b: reverse transmission

There were 117 conquest episodes, coded by independent AI coders blind to content scores. On the 23 double-coded episodes, the coders agreed on adoption in 22 (alpha 0.93), and two disagreements were adjudicated. A source audit of 15 random episodes found the codes fully supported by the cited sources in 10, partly in 5, and contradicted in none. In 55 episodes the conquerors already shared the conquered tradition, and 1 was unknown. Of the remaining 61, adoption within 100 years was full in 5, partial in 16, absent in 39 and unknown in 1.
Primary analysis: with 5 full adoptions, the model was not fitted (minimum 10). The tradition-level permutation test was null: r = +0.04, p = 0.84, 41 episodes.
Partial adoption counted as adoption: odds ratio 0.60 per SD of CCI (0.31 to 1.15; 90 percent interval 0.34 to 1.04), cluster p = 0.13, permutation p = 0.39; 57 episodes, 21 adoptions.
Secondary analysis (CCI difference): not estimable, with 1 adoption among the 33 episodes in which both traditions are scored.
The point estimate is below 1, so H4b is not supported under the Section 14.2 rule. Adoption tracked the conquerors' own tradition: conquerors without a codified religion adopted in 11 of 15 episodes, against 3 of 26 for Christian and Muslim conquerors.

15.5 H9a: moral valence of the seven cooperative behaviours in a new sample

Sample: 30 societies, one per top-level language family, chosen by the fixed rule from 46 verified candidates (Deviation D20). The fixed extraction rule gave 2,368 passages. Coders were independent AI instances not told the hypothesis. On the 474 double-coded passages, they agreed on relevance in 455 (alpha 0.87) and on valence in 452 (alpha 0.86); neither coder recorded a negative judgement in that set, so agreement on negative judgements is untested. Among the 84 passages both coders judged relevant, valence agreement was alpha 0.93. An independent re-derivation from the raw data reproduced every revised-H8, H4b and H9a number reported here.
Relevant passages: 453. Passages with a stated evaluation: 203, of which 200 were positive and 3 negative. By behaviour, positive and negative counts, with the 95 percent Wilson interval for the share negative:

- family: 11 and 0 (0 to 26 percent);
- group: 15 and 0 (0 to 20 percent);
- reciprocity: 35 and 1 (0.5 to 14 percent);
- heroism: 34 and 0 (0 to 10 percent);
- deference: 37 and 1 (0.5 to 13 percent);
- fairness: 18 and 0 (0 to 18 percent);
- property: 50 and 1 (0.3 to 10 percent).

No behaviour exceeds the 5 percent criterion, so H9a is not falsified for any behaviour. No upper bound falls below 5 percent, so the result is consistent with near-universal approval but does not establish it. The negative cases are cattle stolen from enemies (Khoikhoi), a sect's rejection of homage and taxes to the state (Russian 'wanderers'), and a Bedouin's refusal to thank for aid owed as a duty (Lane, Cairo). Using the second coder's codes where they exist leaves every count within 2.

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